User guide

FORGE User Guide

The seeded motif generator on SEQ TOOLS F5



FORGE ONE · Felucca 1.6.1 + FORGE 0.16 · M-VAVE FM-1
github.com/kkasemsa/FORGE-ONE (forge branch) · 24 October 2026
Felucca by Hügelton Instruments and contributors · GPL-3.0-only

FORGE 0.13 in FORGE ONE (Felucca 1.6.1) · M-VAVE FM-1

FORGE writes a musical idea onto a track in one key press, then lets that idea evolve: a melody on a synth track, a beat on a DRUM track (section 6). From 0.6, **STARTER** on the G5 key writes all four tracks at once, sounds included (section 7). FORGE lives in Felucca's SEQ TOOLS layer on the **F5** key. Every pattern comes from a **seed**: the same seed and settings always give the same steps, so you can note a seed you like and get the pattern back later.

All values in this guide are starting points. Judge the result by ear and turn the knobs until it sits right.

1. Quick start

1. Choose a track with **ALGORITHM**. This quick start is for a synth track; a DRUM track makes a beat instead (section 6).
2. Set the track's **ROOT** and **SCALE** on the SCL page, and its length with SEQ TOOLS' **K1 LEN**. FORGE reads all three, so set them first.
3. Press **PLAY** if you want to hear the result as you work.
4. Hold **SEQ** and tap **F5**. A pattern is written at once, and the screen shows it as a grid of steps.
5. Turn **K1 SEED** until you hear an idea you like. Then shape it with **K2 DENSITY**, **K3 EVOLVE** and **K4 STYLE**. Like the rhythm but not the notes? Press **C4** for new notes. Like the notes but not the rhythm? Press **D4**.
6. Press **OCT+** to keep it, or **OCT−** to put the old steps back.

If you change your mind after keeping a pattern, hold **SAVE** to undo the whole FORGE in one step, as long as you have not used another SEQ TOOLS key since.

2. Controls while FORGE is open

Control	What it does	Range
K1 SEED	Picks the pattern. Each seed is a different idea; turning past 99 wraps to 0	0–99
K2 DENSITY	How many of the style's steps sound; above 75 % it also adds notes off the style's rhythm	10–100 %, steps of 5
K3 EVOLVE	How much the idea changes across the pattern and from loop to loop (section 4)	0–4
K4 STYLE	The rhythm and pitch profile (section 3). Choosing a style sets DENSITY to that style's default; turning past AMBIENT wraps to BERLIN	8 styles
C4 (RANDOM)	New notes on the same rhythm. Each press is the next variation	NOTES 0–99
D4 (COOK)	A new rhythm; each step that still sounds keeps its pitch (with EVOLVE, later copies may drift differently)	RHYTHM 0–99
OCT+	Keep the pattern and close FORGE	
OCT−	Cancel: the track's steps go back to how they were before FORGE opened	

Control	What it does	Range
F5 again	Keep the pattern (same as OCT+)	
Another SEQ TOOLS key (not C4 or D4)	Keeps the FORGE pattern, then runs that tool	
Letting go of **SEQ**	Keeps the pattern	
SAVE held (afterwards)	Undo the whole FORGE: the steps and their locks come back (if no other tool ran since)	

Turning the knobs while the sequencer plays is safe: each step is replaced whole, so you never hear half-written steps. Each knob turn rewrites the pattern from the steps as they were before FORGE opened, so turning a knob back returns the earlier pattern exactly. The exception is STYLE: any turn of K4 sets DENSITY to the style's default, so set DENSITY after choosing the style.

What the screen shows. Each time FORGE opens or a knob turns, the header briefly shows `FORGE <seed> <style>`, for example `FORGE 42 BERLIN`. The four knob cards show SEED, DENSITY, EVOLVE and STYLE. Below them, every step of the pattern is a cell, 16 to a row:

- a **bright cell** is a note, with a small mark at its pitch (higher mark, higher note)
- a **half-bright cell** is a tie, holding the note before it
- a **dark cell** is a rest

Under the cells, **C4 NOTES** and **D4 RHYTHM** show how many times you have rerolled each (lit once above 0). Turning K1 SEED or K4 STYLE sets both back to 0.

The footer shows **OCT+ APPLY** and **OCT− CANCEL**. With MENU > SYSTEM > HELP on, a second row shows K1 SEED, K3 EVOLVE and K4 STYLE.

F5 lights up while FORGE is open.

3. The eight styles

Style	Knob card	Motif	Rhythm	Pitch	Octave	Default DENSITY
BERLIN	BERLIN	8 steps	16th ostinato, accents on 1 and 9	Small steps around the root	C3	85 %
DETROIT	DETROIT	16	Syncopated stabs	4-note scale chords	C4	70 %
DUB	DUB	16	Off-beats, long ties	3-note scale chords, narrow range	C3	55 %
MELODIC	MELO	16	8ths	Rises then falls in an arc	C4	75 %
ACID	ACID	16	16ths with accents and slides	Root pedal, other notes nearby, octave jumps	C2	75 %
TRANCE	TRANCE	8	Rolling 16ths	Arpeggios of the scale's triad and octave	C4	90 %
SYNTH	SYNTH	8	8ths	Root, fifth and octave, a third now and then	C3	90 %

Style	Knob card	Motif	Rhythm	Pitch	Octave	Default DENSITY
AMBIENT	AMBI	16	Very sparse, long ties	Wide intervals, some 3-note chords	C4	50 %

- **Motif** is the length of the idea before it repeats. If LEN is shorter than the motif, the motif is LEN steps long.
- **Octave** is where the root sits: the root note is the track's ROOT in that octave (A minor on ACID starts on A2). Melodies move a few scale steps above and below the root (AMBIENT more than an octave). Octave jumps only go up.
- **Step 1 is always the root**, so every pattern starts at home.
- **The scale decides the notes.** Every note is in the track's ROOT and SCALE. Chords, and the TRANCE and SYNTH arpeggios, are built from real intervals (a third, fifth and seventh above the note) kept inside the scale, so they work in every scale. In CHR the chords are minor (root, minor third, fifth, minor seventh); in the pentatonic scales the "seventh" is the scale's sixth; in WHOLE the chords are augmented.
- **The knob card** shortens MELODIC and AMBIENT so they fit. The header shows the full names.

Picking a sound for each style. These are starting points:

Style	Try it on
BERLIN	ANALOG or PHASE with a short decay; a slow LFO on the filter
DETROIT	WHEEL, FM6 or ANALOG pads with a short envelope; a 1/8D delay
DUB	ANALOG or FM6 chords with HALL reverb and a dotted delay; ECHO THROW in the FX layer
MELODIC	FM6, PHASE or TRIO leads
ACID	ANALOG with resonance up, MONO or LEGATO voice mode so the slides glide
TRANCE	ANALOG or TRIO with the gate short; the SLICER for movement
SYNTH	ANALOG or LOFI bass
AMBIENT	GRAIN, PHYS or FM6 pads with long release and HALL reverb

Chords need a polyphonic track: set the voice mode to **POLY** on EDIT > VOICE (K1). On MONO, LEG or UNI only one note of each chord sounds.

The track's sound sets the role (0.12). FORGE reads the category of the selected track's sound (the one the PRESETS list filters by: BASS, LEAD, PAD, PLUCK, KEYS, ...; a FORGE PACK sound has its base preset's) and fits the style to it. The STYLE still sets the genre; the role sets the register and the shape:

Sound	What FORGE writes	The header says
BASS (e.g. ANALOG SQR BASS, SUB BASS, ACID; T1's default)	Single notes, no chords, in the octave of C2, close to the root (MELODIC, TRANCE and AMBIENT walk the root, fifth and octave instead of arcs, arpeggios and wide leaps)	`BASS 42 BERLIN`

Sound	What FORGE writes	The header says
PAD (e.g. ANALOG SOFT PAD, STRINGS, VOICE CHOIR AAH, GRAIN CLOUD PAD)	A chord on the bar's 1 and 9 only, held with ties, moving by a scale step at a time, voiced between F#3 and F4 at the root; no slides, ratchets or octave jumps; EVOLVE only drops chords, never adds them	`PAD 42 BERLIN`
LEAD, PLUCK, KEYS, FX, OTHER	The style as it is (the table above)	`FORGE 42 BERLIN`

To get a style exactly as written on a bass or pad sound, load a LEAD sound, open FORGE, apply, then change the sound: the steps stay. STARTER is not affected: its parts are BASS, LEAD, PAD and DRUM by design.

4. EVOLVE: how the idea changes

FORGE writes the motif once, then copies it through the track's LEN. EVOLVE decides how much the copies differ and how much each loop differs from the last.

EVOLVE	What happens
0	A pure loop. Every copy is the motif, with no chance, conditions or ratchets
1	Later copies sometimes sit a scale step higher or lower, and a few notes move a step. A few ornament notes get a 75 % or 50 % chance. ACID and TRANCE ratchet a few notes
2	More drift. Some ornament notes get a **condition** instead of a chance: 2:2, 3:4, 2:4 or 4:4
3	As 2, and later copies also drop a note or add one in a rest
4	The most change: drift, rhythm changes and ornaments on more notes

- **The first copy keeps the motif's notes and rhythm.** Drift and note changes only happen in later copies, and build up the further along they are. The first copy can still get chances, conditions and (ACID, TRANCE) ratchets.
- **If LEN equals the motif length** (a 16-step style on LEN 16), there is only one copy, so EVOLVE only adds chances, conditions and ratchets. Use a longer LEN, or an 8-step style, to hear the copies drift.
- **Conditions** make one pattern change over several loops: a 4:4 note plays only on every fourth loop, and 2:2 only on every second. Each track counts its own loops from PLAY and from the start of each SONG section.
- **Chance** notes play on some loops and not others, so the line breathes.
- You can see and edit the chances and conditions FORGE wrote on **SEQ > DETAIL** (K1).

For long-form playing, try a short motif on a long LEN: for example BERLIN (8 steps) on LEN 64 at EVOLVE 2. The ostinato stays recognisable while the copies and conditions keep it moving.

5. Recipes

These are starting points. Each one lists the track settings to make first, then the FORGE settings.

Idea	Track settings	FORGE settings
Berlin School sequencer	ANALOG, LEN 32, DIV 1/16, A MIN	BERLIN, DENSITY 85, EVOLVE 2
Dub techno chord stab	FM6 or ANALOG, POLY, LEN 16, D MIN, HALL reverb	DUB, DENSITY 55, EVOLVE 1
Rolling acid line	ANALOG, MONO, LEN 16, resonance up, E MIN	ACID, DENSITY 75, EVOLVE 1, then turn SEED
Melodic techno lead	FM6 or PHASE, LEN 32, F# MIN	MELODIC, DENSITY 70, EVOLVE 2
Trance arpeggio	ANALOG or TRIO, LEN 16, gate short	TRANCE, DENSITY 90, EVOLVE 0
Ambient drift	GRAIN or PHYS, POLY, LEN 64, DIV 1/8, C MAJ	AMBIENT, DENSITY 50, EVOLVE 3

Working with a pattern you like. Note its seed, style and the C4 / D4 numbers. After you apply it, the steps are ordinary Felucca steps: edit single notes on SEQ > STEP, add parameter locks, or use the other SEQ TOOLS (REVERSE, SHIFT, COOK) on top.

6. FORGE on a drum track

From 0.5, F5 works on a DRUM track too. The same four knobs and the same eight style names make a **beat** instead of a melody. The map shows FORGE as a small **F** cell at the right end of the lanes row.

Control	On a DRUM track
K1 SEED	Picks the beat
K2 DENSITY	How many hits. Each style's main hits stay from 50 % up; above 75 % extra 16th hats and rim ghosts come in
K3 EVOLVE	0 is a pure loop. From 1, later bars move a hat or two and add ghost snares and rims with a chance; from 2 the last bar ends in a **snare roll fill on a 4:4 condition** (every 4th loop); from 3 kicks off the beat come and go
K4 STYLE	The beat (table below). Sets DENSITY to the beat's default
C4 (LOW)	New kick, snare, clap and tom; the hats stay
D4 (HATS)	New hats, rim and bell; the low lanes stay

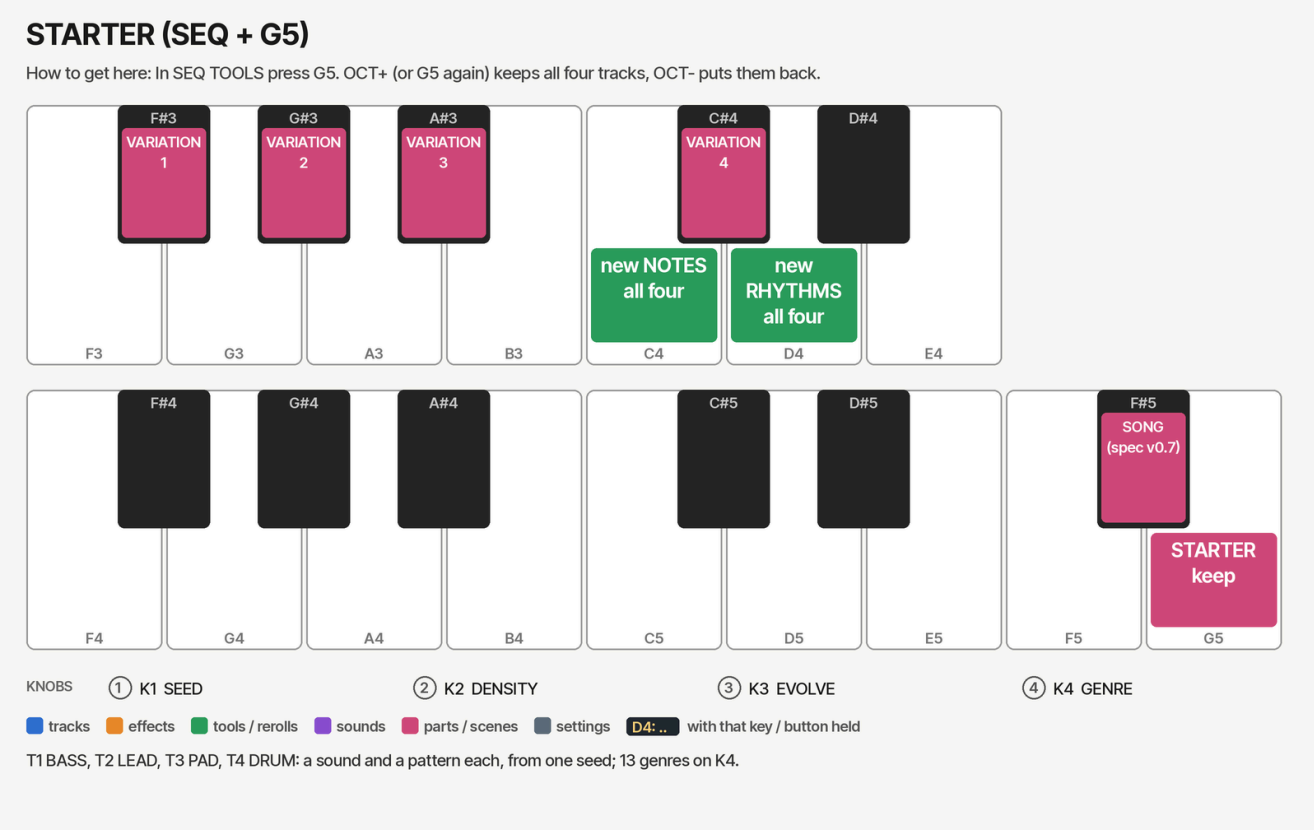
The screen shows every step as a column of eight strips, KICK at the bottom and BELL at the top; a brighter strip is an accent. At LEN 33 to 64 the strips would be too thin, so each step shows two columns of marks instead: kick, snare, clap and tom on the left, hats, rim and bell on the right. The kick on step 1 is always there, and kicks on 1 and 9 are accented; snares and claps on 2 and 4 are accented half the time. Velocity follows the beat as on melodic tracks. Lane mutes (EDIT layer) and lane levels are not touched.

Style	Kick	Snare / clap	Hats	Extras	Default DENSITY
BERLIN	Four on the floor	None	Straight 8ths, 16ths at high density	Rim ticks, an open hat	70 %

Style	Kick	Snare / clap	Hats	Extras	Default DENSITY
DETROIT	Four on the floor, a push on 11	Clap on 2 and 4	8ths, open hat off the beat	Tom, rim	70 %
DUB	1 and 9, a push on 11	Rim on 2 and 4	8ths	Open hat	55 %
MELODIC	Four on the floor	Clap on 2 and 4	Off-beat 8ths, more at high density	Open hat, a tom on 16	70 %
ACID	Four on the floor, a push on 12	Snare on 2 and 4, clap	Off-beat 16ths	Open hat, rim; the most ratchets	75 %
TRANCE	Four on the floor	Clap on 2 and 4	8ths with open hats off the beat	Rim	85 %
SYNTH	1 and 7, a push on 12	Snare on 2 and 4	8ths	Open hat, tom	70 %
AMBIENT	1 only	None	Sparse	Open hat, tom, rim, a bell	45 %

Pick the kit first (PRESETS, or tap **EDIT** on the DRUM track for EDIT 1, then K1 KIT: STD 80 10 66 55 77 DEEP TIGHT PERC 90 MP SD, the 0.14 kits 90 MP SD at the end; PRESETS has no new presets for them): the same beat sounds very different on 80, DEEP or PERC. The beat is ordinary drum-grid steps once applied, so edit it on the grid, mute lanes in the EDIT layer, or run the lane tools over it.

7. STARTER: a song start on all four tracks



STARTER (SEQ TOOLS **G5**, from 0.6) is FORGE for the whole song. One press loads a sound on each of the four tracks and writes a pattern for each, all from one seed:

Track	Part
T1	BASS
T2	LEAD
T3	PAD (chords)
T4	DRUM (T4 becomes a DRUM track)

1. Select a track and set its **ROOT** on the SCL page. All four tracks take that ROOT. Leave SCALE on CHR to use the starter's own scale, or set one to use yours.
2. Hold **SEQ** and tap **G5** (on any track). Four sounds load and four patterns are written at once; the tempo goes to the starter's BPM (not when the FM-1 follows an external MIDI clock).
3. Turn **K4 GENRE** to choose one of 13 genres, and press **black keys 1–4** (F#3 G#3 A#3 C#4) to choose the genre's variation (3 or 4 each, 45 in all; the screen shows them as a row of cells, the keys light up). Each loads its own four sounds, tempo, scale and swing. Turn **K1 SEED** for another song idea in the same starter. **K2 DENSITY** makes all four busier or sparser (50–150 % of each part's own), and **K3 EVOLVE** works as in FORGE. **C4** gives all four new notes, **D4** new rhythms.
4. Press **OCT+** (or G5 again) to keep it, or **OCT−** to put all four tracks back exactly as they were: sounds, settings, steps, automation and tempo.

The screen shows a row a track: its preset and its first 16 steps. Under them, the C4 and D4 counts.

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
Berlin	1 **SEQ**	*BERLIN SEQ*	TRIO ARP LEAD	*BERLIN PAD*	DRUM DEEP KIT · Berlin	120	MIN	–
	2 **DRIFT**	LOFI WAVE BASS	*BERLIN RESO*	*STRING MACH*	DRUM PERC KIT · Ambient	108	MIN	–
	3 **CHOIR**	FM6 FM BASS	TRIO SYNC LEAD	*CHOIR PAD*	DRUM DEEP KIT · Berlin	116	HARM	–
	4 **ARP**	*DEEP BASS*	LOFI STEP LEAD	*BERLIN PAD*	*RHYTHM BOX* · Berlin	124	DOR	–
Detroit	1 **STABS**	FM6 FM BASS	*WARM STAB*	FM6 TINE EP	*HOUSE KIT* · Detroit	126	DOR	–
	2 **STRNG**	*DEEP BASS*	ANALOG STRINGS	*HOUSE PIANO*	*HOUSE KIT* · Detroit	128	MIN	–
	3 **ELECT**	*ELECTRO BASS*	LOFI PULSE LD	PHASE STRING	*808 KIT* · Electro	125	PHRY	–
	4 **DEEP**	FM6 FM BASS	*GLASS ARP*	ANALOG SOFT PAD	DRUM DEEP KIT · Melodic	124	MIN	–
Dub Techno	1 **CHORD**	*DUB SUB*	*DUB CHORD*	GRAIN SHIMMER	*DUB KIT* · Dub	118	DOR	–

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
	2 **DEEP**	VOICE WOW BASS	FM6 PAD	GRAIN CLOUD PAD	*DUB KIT* · Dub	120	MIN	–
	3 **MINI**	*DUB SUB*	PHASE RESO	*DUB PAD*	*909 KIT* · Dub	124	DOR	–
	4 **ECHO**	ANALOG SUB BASS	FM6 TINE EP	GRAIN FROZEN	*DUB KIT* · Dub	116	MIN	–
Melodic	1 **ARC**	TRIO FAT BASS	ANALOG SAW LEAD	FM6 PAD	DRUM DRUM KIT · Melodic	122	MIN	–
	2 **ARP**	*DEEP BASS*	TRIO SYNC LEAD	*TRANCE PAD*	DRUM TIGHT KIT · Melodic	124	MIN	–
	3 **PIANO**	FM6 FM BASS	SAMPLE PIANO	GRAIN CLOUD PAD	DRUM DEEP KIT · Melodic	122	DOR	–
	4 **PLUCK**	TRIO FAT BASS	*TRANCE PLUCK*	*CHOIR PAD*	DRUM DRUM KIT · Melodic	123	MIN	–
Acid House	1 **CLASS**	*ACID 303*	*HOUSE PIANO*	ANALOG SOFT PAD	*HOUSE KIT* · Deep house	124	MIN	10%
	2 **JACK**	*ACID SQUARE*	ANALOG RAVE	*CHOIR AAH*	*HOUSE KIT* · Deep house	122	DOR	15%
	3 **FLOOR**	*ACID 303*	LOFI ARP 8BIT	PHASE ORGAN	DRUM DRUM KIT · Acid	128	MIN	–
Acid Techno	1 **ROLL**	ANALOG ACID	*CHIP ARP*	ANALOG SOFT PAD	*909 KIT* · Acid	128	MIN	–
	2 **WARE**	*ACID SQUARE*	LOFI STEP LEAD	ANALOG PWM STR	*909 KIT* · Acid	135	MIN	–
	3 **HARD**	*ACID 303*	FM6 BRASS	PHASE WIRE	*909 KIT* · Acid	140	PHRY	–
Trance	1 **GATE**	ANALOG SUB BASS	TRIO ARP LEAD	*TRANCE PAD*	*HOUSE KIT* · Trance	138	MIN	–
	2 **LIFT**	ANALOG SUB BASS	*SUPERSAW*	*CHOIR PAD*	*HOUSE KIT* · Trance	138	MIN	–
	3 **PSY**	ANALOG SQR BASS	TRIO RING BELL	PHASE RESO	DRUM TIGHT KIT · Trance	142	PHRY	–
Deep House	1 **KEYS**	*HOUSE BASS*	FM6 TINE EP	ANALOG SOFT PAD	DRUM DEEP KIT · Deep house	122	DOR	15%
	2 **ORGAN**	VOICE WOW BASS	*HOUSE ORGAN*	ANALOG STRINGS	*HOUSE KIT* · Deep house	120	MIN	20%

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
	3 **VOCAL**	ANALOG SUB BASS	*VOX CHANT*	FM6 PAD	DRUM DEEP KIT · Deep house	121	MIN	10%
Synthwave	1 **DRIVE**	*OCTAVE BASS*	ANALOG SAW LEAD	ANALOG PWM STR	*SIMMONS KIT* · Synthwave	110	MIN	–
	2 **NIGHT**	TRIO FAT BASS	*BRASS STAB*	FM6 PAD	*SIMMONS KIT* · Synthwave	98	MIN	–
	3 **8BIT**	LOFI WAVE BASS	*8BIT LEAD*	TRIO CHIP CHOIR	DRUM TIGHT KIT · Synthwave	118	MAJ	–
Breakbeat	1 **AMEN**	*REESE*	FM6 BRASS	ANALOG STRINGS	*BREAK KIT* · Breakbeat	134	MIN	10%
	2 **NUSK**	*ACID SQUARE*	LOFI STEP LEAD	PHASE STRING	*BREAK KIT* · Breakbeat	135	PHRY	5%
	3 **ELECT**	*ELECTRO BASS*	*SCREAM*	FM6 PAD	*808 KIT* · Electro	128	MIN	5%
Ambient	1 **CLOUD**	*DRONE BASS*	FM6 BELL	GRAIN CLOUD PAD	DRUM PERC KIT · Ambient	80	LYD	–
	2 **DRONE**	PHYS DRONE STRING	SAMPLE FLUTE	*AIR PAD*	DRUM PERC KIT · Ambient	72	DOR	–
	3 **GLASS**	*DRONE BASS*	PHYS MARIMBA	*CHOIR PAD*	DRUM PERC KIT · Ambient	84	LYD	–
	4 **RAIN**	FM6 FM BASS	TRIO RING BELL	*GLASS PAD*	DRUM PERC KIT · Ambient	76	MIN	–
Lo-Fi	1 **PIANO**	*LOFI BASS*	SAMPLE PIANO	*LOFI PAD*	*808 KIT* · Lo-fi	84	DOR	25%
	2 **TAPE**	TRIO FAT BASS	*WARM KEYS*	WHEEL SOFT FLUTE	DRUM DRUM KIT · Lo-fi	88	MIN	20%
	3 **SAX**	*LOFI BASS*	SAMPLE SAX	FM6 PAD	DRUM PERC KIT · Lo-fi	80	DOR	22%
Jarre	1 **OXY**	ANALOG SQR BASS	*OXY LEAD*	*STRING MACH*	*MINI KIT* · Rhythm box	104	MIN	–
	2 **EQUI**	LOFI WAVE BASS	TRIO ARP LEAD	*CHOIR AAH*	*MINI KIT* · Electro	120	MIN	–
	3 **FIELD**	FM6 FM BASS	*BRASS STAB*	ANALOG PWM STR	*MINI KIT* · Rhythm box	112	DOR	–

Genre (K4)	Black key, variation	T1 bass	T2 lead	T3 pad	T4 kit · beat	BPM	Scale*	Swing
	4 **ZOO**	VOICE WOW BASS	*VOX CHANT*	GRAIN SHIMMER	DRUM TIGHT KIT · Electro	118	PHRY	–

* The starter's SCALE is used only when the selected track's SCALE is CHR (the default). Any other SCALE you set is kept.

FORGE PACK (0.9). Sounds in *italics* in the table are from the FORGE PACK: 43 sounds (0.14) made for the starters. Each is a factory preset with its own settings on top (oscillator and filter, envelope, voice mode and glide, unison, the effect sends, the LFO), exactly as if you had turned the knobs yourself. So PRESETS shows the factory preset's name, a project saves them like any edited sound, and they open on stock Felucca and in the Felucca web editor. To keep one for other songs, save it with **SAVE > USER** while the track has it.

Sound	Built on	Character
BERLIN SEQ	ANALOG SQR BASS	Resonant square sequence, short decay, a little delay
DEEP BASS	ANALOG SUB BASS	Two detuned saws, legato with a touch of glide
DUB SUB	ANALOG SUB BASS	Round sine sub with a little drive
ACID 303	ANALOG ACID	Saw, high resonance, strong filter envelope, slide, driven
ACID SQUARE	ANALOG ACID	The square-wave acid line
HOUSE BASS	TRIO FAT BASS	Short three-pulse bass
OCTAVE BASS	ANALOG SQR BASS	Bright saw for octave 16ths
REESE	TRIO FAT BASS	Detuned three-saw bass with a slow filter wobble
DRONE BASS	ANALOG SUB BASS	Slow-swelling triangle drone, reverb, slow filter drift
LOFI BASS	ANALOG SUB BASS	Warm, dull triangle bass
ELECTRO BASS	LOFI WAVE BASS	8-bit pulse bass, tight
SUPERSAW	ANALOG SAW LEAD	Unison saws, wide spread, chorus, delay and reverb
TRANCE PLUCK	ANALOG PLUCK	Bright saw pluck with delay
OXY LEAD	ANALOG SAW LEAD	Slow PWM lead with glide and vibrato, long reverb
SCREAM	TRIO SYNC LEAD	Hard sync lead with drive
BERLIN RESO	PHASE RESO	Phase-distortion resonant blip with delay

Sound	Built on	Character
BRASS STAB	ANALOG BRASS	Saw brass with a filter swell
DUB CHORD	ANALOG PLUCK	Short filtered chord for dub, heavy delay and reverb
WARM STAB	ANALOG PWM STR	PWM chord stab, chorus
HOUSE PIANO	SAMPLE PIANO	The piano sample, brighter, a little drive
HOUSE ORGAN	WHEEL JAZZ PERC	Percussive organ, fast rotor
GLASS ARP	FM6 BELL	FM bell with delay and reverb
8BIT LEAD	LOFI PULSE LD	8-bit pulse lead with vibrato
CHIP ARP	LOFI ARP 8BIT	8-bit minor arpeggio
VOX CHANT	VOICE VOX LEAD	Formant voice lead with glide
WARM KEYS	FM6 TINE EP	FM electric piano with chorus and tape wobble
STRING MACH	ANALOG STRINGS	Ensemble strings: detuned saws, heavy chorus · LEVEL −8 dB
BERLIN PAD	ANALOG SOFT PAD	Slow pad whose filter opens with the envelope · LEVEL −7 dB
TRANCE PAD	ANALOG PWM STR	Wide PWM pad · LEVEL −7 dB
DUB PAD	ANALOG SOFT PAD	Soft pad with delay · LEVEL −7 dB
CHOIR PAD	VOICE CHOIR AAH	Formant choir, slow · LEVEL −12 dB
CHOIR AAH	VOICE CHOIR AAH	The factory choir as it is, at LEVEL −12 dB (0.9.2)
GLASS PAD	GRAIN SHIMMER	Granular shimmer, long · LEVEL −7 dB
AIR PAD	NOISE WIND	Filtered noise, slowly drifting · LEVEL −7 dB
LOFI PAD	ANALOG SOFT PAD	Soft pad with hiss and tape wobble · LEVEL −7 dB
808 KIT	DRUM DEEP KIT	KIT 80: long, round kick; deep and boomy (0.14; was BOOM KIT)
HOUSE KIT	DRUM DRUM KIT	Punchy kit, snappy (0.14: on KIT 90)
BREAK KIT	DRUM TIGHT KIT	Bright, snappy, driven
RHYTHM BOX	DRUM PERC KIT	An early rhythm machine, soft
909 KIT	DRUM TIGHT KIT	KIT 90: the later analog drum machine, a click on the kick, bright hats (0.14; was TECHNO KIT)
MINI KIT	DRUM PERC KIT	KIT MP: the 70s organ-top rhythm box, soft: bongo, claves, maracas (0.14)
SIMMONS KIT	DRUM DEEP KIT	KIT SD: the 80s electronic kit, swept kick and big bent toms (0.14)

Sound	Built on	Character
DUB KIT	DRUM DEEP KIT	Deep kit with delay and reverb on the drums

These were designed without listening on the FM-1: every starter is checked to play all four tracks without silence or clipping, but the balance between tracks (some leads are louder than their bass) is for your ears. Tell me which sounds to change.

What STARTER changes: each track's sound (engine and factory preset, with its effect sends), its pattern, LEN, DIV (1/16), ROOT and SCALE, and the tempo. Since 0.9.2 a pack pad also sets its track's LEVEL (the pads above; a sustained chord peaks far above a bass line, and CHOIR AAH was too loud on the instrument). **It leaves alone:** PAN and MUTE, LEVEL on every other sound, the master and the effects' own settings. Steps that held something before and are now empty lose their parameter locks when you apply, as with FORGE; a sound load also drops automation of the old engine's own controls.

Undo: STARTER changes four tracks, and SAVE-held undo holds only one, so after STARTER it has nothing to undo. **OCT−** while STARTER is open is the way back. Save your project first if it matters.

Seeds: the same SEED, C4 and D4 numbers, STARTER, DENSITY, EVOLVE and ROOT (and SCALE, if you set one) always give the same four patterns. A set of starter seeds is pinned by a test, as FORGE's are.

CPU: each starter's four engines were chosen to play together, but FORGE ONE has not yet been tested on a real FM-1. If you hear voices drop out, swap the pad for a lighter engine (ANALOG or FM6) on the EDIT layer.

SONG (0.13): four parts and the arrangement from one seed. With STARTER open, press **F#5** (the last black key). SONG uses the starter's sounds, genre, SEED, DENSITY and EVOLVE (the knobs work as in STARTER) and makes four related parts:

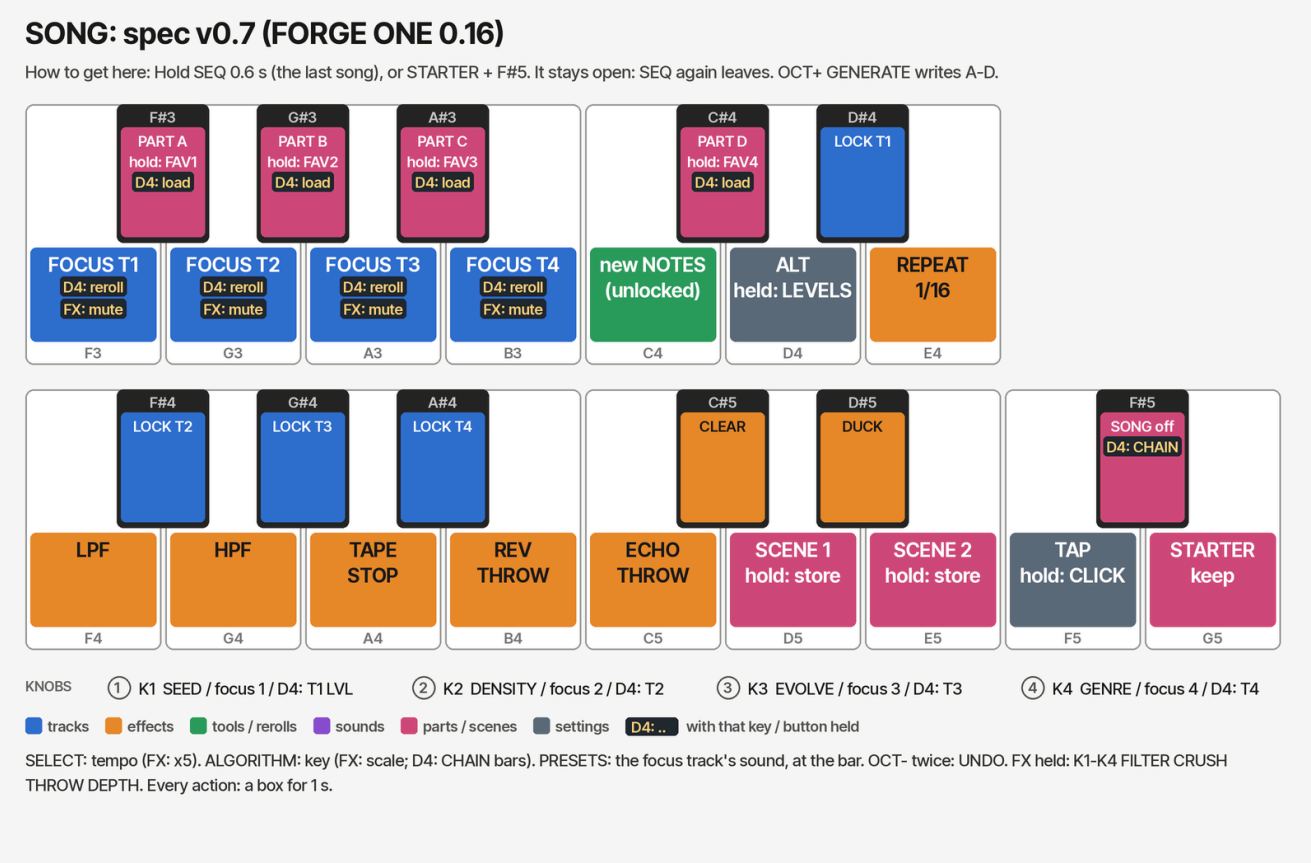
Part (slot)	What it is
A VERSE	The starter as it is, a little sparser (DENSITY ×0.85, the lead ×0.7)
B CHORUS	Denser (×1.15), the lead an octave up and developed (EVOLVE +1), open hats on the off-beat 8ths, step 1 accented
C BRIDGE	A new motif, the bass and pad moved to the scale's 4th or 6th degree, a half-time beat (kick on 1, snare on 9)
D LIFT	(BERLIN, DEEP HOUSE, SYNTHWAVE, AMBIENT, JARRE) the last chorus bigger: lead and pad an octave up, DENSITY ×1.25, 8th hats, a fill on the last bar (AMBIENT: gentler, no hats or fill)
D BREAK	(the other 8 genres) the breakdown: no bass, no kick or snare, sparse hats, a snare roll building over the last bar, the pad's filter down and opening on the last bar

- **Black keys 1–4** play part A–D (its steps go into the four tracks; the selected part is lit). **Black keys 5–8** lock T1–T4. **C4** rerolls the notes of the part shown and **F3–B3** one track, leaving locked tracks exactly as they are. **D4** is FOCUS (below).
- **OCT+ GENERATE** asks `OVERWRITE A-D?` and lists what is in the four project slots now (`--` for an empty one). **OCT+** writes the four parts to slots A–D, named `S42 VERSE`, `S42 CHORUS`, `S42 BRIDGE`, `S42 LIFT` or `S42 BREAK`, and writes the arrangement into GLO > SONG. **OCT−** cancels: no slot is touched and the four tracks go back to how they were before STARTER. The transport must be stopped to save (`STOP TO SAVE`).

- After writing, a reroll asks `SAVE B?` (for the part you rerolled): **OCT+** saves that slot only, **OCT−** keeps the change in the tracks without saving. OCT+ or OCT− otherwise closes SONG, leaving the part shown in the tracks. F#5 again goes back to plain STARTER.
- The arrangement (8 rows, edit it in GLO > SONG):

Row	Section	T1 BASS	T2 LEAD	T3 PAD	T4 DRUM	Repeats (LIFT / BREAK genres)
1	INTRO	-	-	A	A	2
2	VERSE	A	A	A	A	4
3	CHORUS	B	B	B	B	4
4	VERSE	A	A	A	A	4
5	BRIDGE	C	C	C	C	2
6	CHORUS / BREAK	B / D	B / D	B / D	B / D	4 / 2
7	LIFT / CHORUS	D / B	D / B	D / B	D / B	4
8	OUTRO	-	-	B	B	2

The parts are ordinary projects and the arrangement an ordinary SONG chain, so stock Felucca 1.6.1 loads and plays them too. Edit a part as usual: load its slot, change it, save it back to the same slot.



Into SONG in one step (0.16, spec v0.7): hold **SEQ for 0.6 s** (from anywhere, stopped or playing): SONG opens straight away with the last song (its SEED, genre, key, scale, tempo, part, locks and rerolls; kept in the settings when SONG is left or written), skipping the STARTER screen. A short SEQ tap is unchanged, and a key or knob touched while SEQ is held cancels it. Entering SONG by any way locks the layer open, so SEQ

need not be held: to leave, press SEQ again (or HOME, or any page button). **MENU > SYSTEM > BOOT** `SONG` opens SONG with the last song at power-on, stopped, on part A (`NORMAL`, the default, does not).

SONG's keys (spec v0.6, FORGE ONE 0.15; 0.16: spec v0.7). The columns: the key alone, with D4 held, with FX held.

Key	Alone	D4 held	FX held
Black 1–4 (F#3 G#3 A#3 C#4)	Tap: part A–D (playing: at the next bar); **held 0.5 s: store FAV 1–4** (the part stays)	Load FAV 1–4 (at the bar)	Part A–D (D4 + FX: store FAV 1–4)
Black 5–8 (D#4 F#4 G#4 A#4)	Lock T1–T4	same	same
Black 9 (C#5)	CLEAR: every effect, latch, mute and DUCK off	same	same
Black 10 (D#5)	DUCK on / off (non-drum tracks pump on the kick)	same	same
Black 11 (F#5)	SONG on / off	**CHAIN on / off**	same
F3 G3 A3 B3	**FOCUS T1–T4** (the same key again: off)	Reroll T1 / T2 / T3 notes, T4 drum rhythm	Mute T1–T4 while held
C4	Reroll notes, every unlocked track	same	same
D4	ALT: held alone, K1–K4 are T1–T4 LEVEL, a mixer (a tap does nothing)	–	–
E4 F4 G4 A4 B4 C5	Held: REPEAT 1/16, LPF, HPF, TAPE STOP, REV THROW, ECHO THROW (no FX button needed)	same	same
D5 / E5	Tap: recall SCENE 1 / 2 at the next 1/16; **held 0.5 s: store**	–	Store SCENE 1 / 2
F5	**TAP tempo** (the last 4 taps; 2 s starts over); **held 0.5 s: CLICK on / off**	same	same
G5	Applies STARTER, leaves SONG	same	same
SELECT	**Tempo** ±1 BPM	same	±5 BPM
ALGORITHM	**Key** ±1 semitone, at the next bar	**CHAIN bars** of the part shown (1 2 4 8 16)	**Scale**, at the next bar
OCT−	Cancel a change waiting; KEEP; cancel / done (0.35 s later)	–	–
OCT− twice (within 0.35 s)	**UNDO** the last reroll, preset, favourite or KEY / SCALE change (one level)	–	–
PRESETS	**The focus track's preset**, loaded at the next bar (OCT−: cancel)	same	nothing
KNOB 1–4	SEED DENSITY EVOLVE GENRE; FOCUS on: the track's four (below)	T1–T4 LEVEL (they stay when D4 is let go)	FX macros FILTER CRUSH THROW DEPTH (back to 0 when FX is let go)

- A locked track ignores its own reroll key (`T3 LOCKED`). Before the song is written a per-track reroll says `NEW T3`; after, it asks `SAVE X?` like C4. (0.13 / spec v0.5: D4's reroll of every track's rhythm is gone; D4 + B3 rerolls T4's.)
- What F3–B3 do is decided when the key goes down: FX held then, a mute until the key is let go (even if FX is let go first); else D4 held, a reroll; else FOCUS. Pressing D4 or FX after the key changes nothing. Mutes are never saved and never change the chain: a track the chain row silences (`-`, as in INTRO and OUTRO) stays silent.
- With MENU > FX LATCH ON the effects and mutes toggle and stay on across part changes and rerolls; CLEAR turns them off. Leaving SONG (F#5, G5, pressing SEQ again) turns every effect, latch, mute and DUCK off.
- **UNDO (0.16):** press OCT− twice quickly. It takes back the last reroll (C4 or one track), the last preset or favourite loaded, or the last KEY / SCALE change (one level: then `NOTHING TO UNDO`). A single OCT− still does what it did, 0.35 s later (it waits to see if a second press comes).
- **CLICK (0.16):** hold F5 for 0.5 s. A visual metronome: `CLK` in the header and the BAR row flashes on every beat (filled on beat 1, framed on the others). There is no audible click (the FM-1 has only the main output).
- **CHAIN (0.16):** D4 + F#5 turns the auto-chain on: A → B → C → D → A… each part for its bars, changing at the bar, forever. D4 + ALGORITHM sets the bars of the part shown (1, 2, 4, 8 or 16; 8 at first). The state line shows `CHAIN 3/8` (the bar in the part / its bars). Tapping a part while it runs goes there at the bar and the chain carries on from it. It plays the tracks as SONG shows them (not GLO > SONG's arrangement).

FOCUS, faders and scenes (spec v0.5, v0.6):

- **FOCUS:** tap F3 / G3 / A3 / B3 to put T1 BASS / T2 LEAD / T3 PAD / T4 DRUM on the knobs; the knob cards and the track row show it (`FOCUS`). Tap the same key again to turn it off. A locked track can be focused (a lock is for notes and rhythm).
- **Levels (0.15):** hold D4 alone and turn K1–K4: T1–T4 LEVEL, a mixer, FOCUS on or off (0.16: `MIXER` large across the top while D4 is held, the cards show the four levels). Let go: the levels stay, the knobs go back. FX held always wins: its macros, which snap back.

Track	KNOB 1	KNOB 2	KNOB 3	KNOB 4
T1 BASS	CUTOFF	RESO	ENV MOD (FLT)	DECAY
T2 LEAD, T3 PAD	FILTER	A/R (attack and release together)	SEND (REV and DLY together)	LEVEL
T4 DRUM	KICK level	TUNE (the kit's)	DECY (the kit's)	DRV

The filter and RESO are the engine's own (CUT, FREQ, TONE, DCW..; RES or Q, else FB, FOLD or DUTY, else the track's DIST). The knobs are encoders, so a value always moves on from where it is: nothing jumps when you change track or recall a scene. These are the tracks' real settings: they stay across parts, rerolls and CLEAR, and are saved with a part like any sound edit. - **Scenes:** hold D5 / E5 for 0.5 s (or FX + D5 / E5) to store the knob settings of all four tracks; tapping D5 / E5 brings them back on the next 1/16 (stopped: at once). An empty one says `SCENE 2 EMPTY`. Scenes live until power-off and are not saved with the song. A track whose sound changed since (engine, preset) is skipped: the box says `T2 SKIPPED (SOUND)`.

Tempo, key and sounds without a hold (spec v0.6): **SELECT** sets the tempo (FX held: 5 at a time), **F5** taps it; BPM LOCK and an external clock refuse both. **ALGORITHM** moves the key a semitone, with FX the

scale (MIN DOR PHRY MPEN HARM MAJ LYD MIX PEN); playing, it waits for the next bar (the key shows in the accent until then), then all four tracks are written again in it. **PRESETS** browses the focus track's presets (FOCUS off: the last focused, T1 at first); the sound changes at the next bar at least 0.6 s after the last turn (stopped: 0.6 s after it), and OCT− before then cancels it. **Favourites:** D4 + FX + black 1–4 stores the focus track's sound in FAV 1–4 (a factory preset, or the user preset it plays; kept in the settings), D4 + black 1–4 loads it at the next bar (a DRUM kit only on T4, and only a kit there).

SONG's screen (spec v0.6): read at arm's length. At the top the tempo and the key; the parts A–D (the one playing filled); `BAR 3/8` and `STEP 09/16`; a bar for where the part (or GLO > SONG's chain) is; 16 cells with the playhead; the four tracks with a dot that lights on each note, a level meter, `M` for a mute, `L` for a lock and `FOCUS`; the two scenes and the state or the last message; the four knob cards as the knobs are now.

Every action shows (0.16, spec v0.7 §A): each key, knob or encoder in SONG opens a box in the middle of the screen for about a second: what it is small (`T1 BASS`, `PART`, `FAV 2 SAVED`, `TEMPO`), what happened large (`CUT 74`, `B`, the preset's name, `126 BPM`), and a bar for a value or `@ NEXT BAR` for a change that waits. The next action replaces it at once. Something refused (`BPM LOCKED`, `EXT CLOCK`, `T3 LOCKED`, `FAV 2 EMPTY`) has a red frame. A part that differs from its saved slot has an orange dot on its block (all four until the song is written; `SAVE A-D` or `SAVE X?` clears them).

8. What FORGE writes, and what it leaves alone

Writes: steps 1 to LEN of the selected track. On a synth track each step gets notes (1, 3 or 4), NOTE, TIE or REST, accent, slide, ratchet (×2 or ×3), and a chance or condition. Velocity follows the beat: notes on the beat are louder and off-beat 16ths softer, with a little variation, between 40 and 124 (BERLIN, TRANCE and ACID the widest, DUB and AMBIENT the gentlest). Accented notes play at 127 as always. Every copy of the motif has the same velocities.

Leaves alone:

- steps past LEN
- the track's LEN, DIV, SWING and GATE, its sound, and its automation
- the other three tracks

Parameter locks: when you apply a pattern, steps that held something before and are now empty lose their locks, as with EUCLID. Undo (SAVE held) brings them back.

Compatibility: FORGE only uses fields that Felucca 1.6 already stores, so projects made with FORGE open on stock Felucca 1.6 and 1.6.1. Projects with conditional steps do not open on Felucca 1.5.1 or older. FORGE at EVOLVE 2 and up can write conditions, so check SEQ > DETAIL or AUTOMATION before going back.

Seeds and settings: the same SEED, C4 and D4 numbers, STYLE, DENSITY, EVOLVE, LEN, ROOT and SCALE always give the same steps. From FORGE 0.3 on, a set of seeds is pinned by a test, so a later version only gives a seed a different pattern when its changelog says so. Patterns you want to keep for certain belong in a project. Change any of them and the pattern changes. At EVOLVE 0, changing DENSITY keeps the pitch of the notes that stay; with EVOLVE 1 and up it can also change how the later copies drift.

9. When it does not do what you expect

What you see	Why, and what to do
F5 does nothing	A SONG is playing ("STOP TO EDIT"), or the firmware is older than 0.5 on a DRUM track
G5 does nothing	A SONG is playing ("STOP TO EDIT"), or the firmware is older than 0.6
STARTER changed my sounds and I want them back	Press **OCT−** before letting go of SEQ. After applying, there is no undo for STARTER: load your project again
The notes are too low or too high	Each style has its own octave. To move a pattern, set **QNT** to **SEQ+T** on the SCL page and turn **TRN** (TRN alone, and the octave buttons, only move the keys). Or try a style in another octave
Chords play as single notes	The voice mode is MONO, LEG or UNI. Set **POLY** on EDIT > VOICE
ACID slides do not glide	Use MONO or LEG voice mode, and some GLIDE on EDIT > VOICE (K2)
The pattern changes every loop and I want it fixed	Set EVOLVE to 0, or remove chances and conditions on SEQ > DETAIL
The knobs edit LEN, not SEED	FORGE is not open (F5 is not lit). Tap F5 while holding SEQ
I applied it and want the old steps back	Hold **SAVE** before using another tool (one undo). Or press **OCT−** before letting go of SEQ: it puts back everything the layer changed since it opened, LEN, DIV, SWING and GATE included

10. Credits

FORGE is part of FORGE ONE, a fork of Felucca by Leo Kuroshita, Hügelton Instruments, and contributors. GPL-3.0-only. Source: github.com/kkasemsa/FORGE-ONE, `forge` branch; FORGE's code is in `firmware/src/ui_forge.c`, STARTER's in `firmware/src/ui_starter.c`.